

The Looking-Glass Transform: The Schrödinger Equation as a Centered-Difference Approximation of the Classical Liouville Equation

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Abstract

The Schrödinger equation lies a single approximation step from the classical Liouville equation. Lifting classical Hamiltonian dynamics to a conservation law for the phase-space distribution, Fourier transforming in momentum, and applying a centered-difference approximation to the potential gradient yields an equation that separates into coupled Schrödinger equations; the phase-space distribution is the Wigner quasi-probability function, and the Born position density is its momentum marginal. The ingredients of this route are classical results of the phase-space formulation of quantum mechanics, and closely related Liouville-to-Schrödinger constructions exist. The contribution here is the framing and its consequences: the centered difference is kept finite rather than taken as a limit, so quantum dynamics appears as a controlled discretization of classical dynamics—exact for quadratic potentials, with explicit leading error $(\hbar^2/24) \partial^3 V / \partial x^3 \partial^3 w / \partial p^3$, the Moyal correction read in reverse. Two observations follow from this viewpoint: Wigner negativity admits a classical reading as fluctuation below a uniform background, since $w \rightarrow w + c$ is invisible to the Liouville flux; and the transformed dynamics is well posed on a state space strictly larger than the quantum one, which contains the classical point preparations excluded by the uncertainty principle. For a separated entangled pair measured by an apparatus that interacts only with the distant particle, every step of the transform is exact, and collapse is conditioning of the signed joint distribution on the pointer marginal: the flow is local, the Born weights emerge as the apparatus grows, and—unlike classical conditioning—conditioning can create negativity. We are explicit about what the framing does not do: where the approximation error accumulates—anharmonic potentials at long times—experiment follows the quantum dynamics, not the classical one. The value of the transform is

structural and pedagogical: it locates exactly what must be added to probabilistic classical mechanics to obtain quantum mechanics, in the language of Fourier transforms and finite differences.

1 Introduction

Quantum mechanics works extraordinarily well, yet its interpretation remains contested. Why does the wavefunction give only probabilities? What happens during measurement? Does the universe branch into many worlds to account for each outcome [1]? These questions persist in part because the standard formulation takes the Schrödinger equation and the Born rule as postulates.

The phase-space formulation of quantum mechanics [2, 3, 4, 5] makes the kinship with classical mechanics most visible: the quantum evolution of the Wigner function differs from the classical Liouville flow only by terms of order $\hbar^2$, and is identical to it for quadratic potentials [6]. That correspondence is usually read from quantum to classical: start with a wavefunction, construct the Wigner function, and recover classical mechanics as $\hbar \rightarrow 0$.

This paper reads it in the other direction. We start from classical Hamiltonian mechanics with probabilistic initial data and apply three elementary steps: lift the equations of motion to a conservation law for the phase-space distribution, Fourier transform in momentum, and apply a centered-difference approximation to the potential gradient. The result separates into coupled Schrödinger equations. We call the composite the looking-glass transform.

The reverse direction has been explored before. Carnovali and França [7] derive a Schrödinger-type equation from the Liouville equation by an equivalent route, taking the mean-value-theorem limit in which the difference step becomes exact. Bondar et al. [8] obtain both quantum and classical (Koopman–von Neumann [9]) dynamics from operational axioms, with the Wigner function itself serving as the wave function. Hall, Deckert, and Wiseman [10] build quantum behavior from many interacting classical worlds by entirely different machinery. The present treatment differs in keeping the difference step finite and explicit: quantum dynamics appears as a controlled discretization of classical dynamics, exact for quadratic potentials, with the leading error term stated and retained rather than removed by a limit.

We are explicit about what is and is not new here. The transform itself is not: the route from the Liouville equation through a momentum Fourier transform and shifted variables to coupled Schrödinger equations was published by Carnovali and França [7], and the underlying correspondence is the classical Wigner–Moyal one [6, 5]. What we offer is a particular reading of that route and four observations that follow from it. The first is or-

ganizational rather than novel: the entire difference between the dynamics of probabilistic classical mechanics and quantum mechanics is one explicit term—the Moyal correction read in reverse—retained here as a finite error rather than removed by the limit taken in [7]. The second is a reading of Wigner negativity: a uniform background $w \rightarrow w + c$ is invisible to the Liouville flux, so w behaves like an acoustic pressure perturbation, and its negative values are rarefactions below an ambient level rather than negative probabilities. Negativity is usually treated as the irreducibly quantum feature of w [11]; Section 8.4 argues that for a model whose predictions—its marginals—are non-negative and correct, it does not matter. The third is structural: the transformed dynamics is well posed on a state space larger than the quantum one—large enough to contain the classical point preparations the uncertainty principle excludes—within which the quantum states, those with positive-semidefinite C , form an invariant subset. The constituent facts here are known (the bounds on valid Wigner distributions are classical [12]); what we add is only the reading of the larger set as a dynamical arena. The fourth concerns the collapse of a separated entangled pair (Section 8.3). For free particles and a von Neumann pointer coupled only to the distant particle, every Hamiltonian is quadratic, so the transform is exact at every step and measurement is classical Liouville flow of a signed distribution followed by conditioning on the pointer marginal. The flow is local, nothing done at the distant particle signals, and the Born weights emerge as the apparatus grows; the quantum ingredient is confined to the sign of the initial distribution, and conditioning with signed weights can create negativity that classical conditioning never could. The underlying identity—that conditioning the joint Wigner function reproduces the post-measurement state—is standard, and Bell made the classical-conditioning point for positive Wigner functions [13]; what we add is the exact, fully classical account of the measurement interaction and its numerical illustration.

Two facts frame the view: the phase-space distribution $w(x, p, t)$ is the Wigner function [2]—a signed field satisfying a local conservation law whose marginals are genuine probabilities—and where the approximation error accumulates, experiment follows the quantum dynamics, not the classical one (Section 6). Within this representation the Born position density is not an additional postulate but the momentum marginal of w ; what this does and does not explain about measurement is taken up in Section 7.

We work in one spatial dimension for clarity; the extension to higher dimensions and multiple particles is straightforward.

2 Quantum Perspective

Consider a tiny one-dimensional non-relativistic particle of mass M in a tiny potential $V(x, t)$. Here the accepted model is the Schrödinger equation, where $|\Psi(x, t)|^2$ gives the probability of observing the particle at location x at time t . The wave function $\Psi(x, t)$ evolves according to the partial differential equation

$$i\hbar\Psi_t = -\frac{\hbar^2}{2M}\Psi_{xx} + V\Psi, \quad (1)$$

where $\hbar \approx 1.05 \times 10^{-34}$ joule-seconds.

3 Classical Perspective

Consider a regular one-dimensional non-relativistic particle of mass M in a regular potential $V(x)$. The position of the particle $x(t)$ and momentum of the particle $p(t)$ evolve according to the ordinary differential equation

$$M\frac{dx}{dt} = p, \quad (2)$$

$$\frac{dp}{dt} = -\frac{\partial V}{\partial x}. \quad (3)$$

4 Three Concepts to Get Quantum Mechanics from Classical Mechanics

What must be added to classical mechanics to get quantum mechanics? We combine the following concepts.

4.1 An ordinary differential equation can be lifted to a linear partial differential equation

Think of an ordinary differential equation initial value problem as evolving a partial differential equation which describes the evolution of the probability distribution of the initial values [14].

ODE:

$$\frac{du_1}{dt} = v_1, \quad (4)$$

$$\frac{du_2}{dt} = v_2, \quad (5)$$

$$\vdots \quad (6)$$

$$\frac{du_n}{dt} = v_n. \quad (7)$$

PDE:

$$\frac{\partial w}{\partial t} + \frac{\partial(v_1 w)}{\partial u_1} + \frac{\partial(v_2 w)}{\partial u_2} + \cdots + \frac{\partial(v_n w)}{\partial u_n} = 0. \quad (8)$$

Using the Dirac delta function for the initial value, the PDE recovers the ODE initial value problem: $w(u, 0) = w_0(u) = \delta(u - u_0)$.

4.2 The Fourier transform

$$\hat{\Psi}(\hat{p}) = \int_{-\infty}^{+\infty} dp e^{-i\frac{\hat{p}p}{\hbar}} \Psi(p), \quad (9)$$

$$\Psi(p) = \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar} e^{+i\frac{\hat{p}p}{\hbar}} \hat{\Psi}(\hat{p}), \quad (10)$$

$$\partial_p \Psi \rightarrow i\frac{\hat{p}}{\hbar} \hat{\Psi}, \quad (11)$$

$$p\Psi \rightarrow i\hbar\partial_{\hat{p}} \hat{\Psi}. \quad (12)$$

$$\delta(\hat{p}) = \int_{-\infty}^{+\infty} \frac{dp}{2\pi\hbar} e^{i\frac{\hat{p}p}{\hbar}}. \quad (13)$$

The use of $\hat{p} = \hbar\omega$ compared to the typical spectral parameter ω greatly compresses the spectrum.

4.3 Centered-difference approximation of $\frac{\partial V}{\partial x}$

From the Taylor expansion of V around x , we have

$$\frac{\partial V}{\partial x} \approx \frac{V(x + \varepsilon/2, t) - V(x - \varepsilon/2, t)}{\varepsilon} + O(\varepsilon^2). \quad (14)$$

We actually approximate $\varepsilon \cdot \frac{\partial V}{\partial x}$, so with first error term:

$$\varepsilon \frac{\partial V}{\partial x} \approx V(x + \varepsilon/2, t) - V(x - \varepsilon/2, t) - \frac{\varepsilon^3}{24} \frac{\partial^3 V(x, t)}{\partial x^3} + O(\varepsilon^5). \quad (15)$$

5 Quantum from Probabilistic Classical Mechanics: The Looking-Glass Transform

We now compose the three concepts. The result reveals the Schrödinger structure *inside* classical mechanics, with the quality of the approximation controlled by the explicit error term of Section 6.

We start with the classical Hamiltonian ordinary differential equation:

$$\frac{dx}{dt} = \frac{1}{M}p, \quad (16)$$

$$\frac{dp}{dt} = -\frac{\partial V}{\partial x}. \quad (17)$$

Using the conservation of probability, this corresponds to the following partial differential equation describing the evolution of the probability density:

$$\frac{\partial w}{\partial t} + \frac{\partial}{\partial x} \left(\frac{1}{M}pw \right) + \frac{\partial}{\partial p} \left(-\frac{\partial V}{\partial x}w \right) = 0, \quad (18)$$

or

$$w_t + \frac{p}{M}w_x - \frac{\partial V}{\partial x}w_p = 0. \quad (19)$$

Now Fourier transform only with respect to the momentum p and multiply by $i\hbar$. This results in

$$i\hbar\hat{w}_t - \frac{\hbar^2}{M}\hat{w}_{x\hat{p}} + \hat{p}\frac{\partial V}{\partial x}\hat{w} = 0. \quad (20)$$

We now change variables. Define

$$z_{\pm} = x \pm \frac{\hat{p}}{2}, \quad (21)$$

So

$$\frac{\partial}{\partial x} \rightarrow \frac{\partial}{\partial z_+} + \frac{\partial}{\partial z_-}, \quad (22)$$

$$\frac{\partial}{\partial \hat{p}} \rightarrow \frac{1}{2}\frac{\partial}{\partial z_+} - \frac{1}{2}\frac{\partial}{\partial z_-} \quad (23)$$

This transforms (20) into

$$i\hbar\hat{w}_t - \frac{\hbar^2}{2M} \left[\left(\frac{\partial}{\partial z_+} \right)^2 - \left(\frac{\partial}{\partial z_-} \right)^2 \right] \hat{w} + \hat{p}\frac{\partial V}{\partial x}\hat{w} = 0 \quad (24)$$

Next is the only approximation. We use (14) with $\varepsilon = \hat{p}$:

$$\hat{p} \frac{\partial V}{\partial x} \approx V(x + \hat{p}/2, t) - V(x - \hat{p}/2, t) \quad (25)$$

This transforms (20) into

$$i\hbar \hat{w}_t - \frac{\hbar^2}{2M} \left[\left(\frac{\partial}{\partial z_+} \right)^2 - \left(\frac{\partial}{\partial z_-} \right)^2 \right] \hat{w} + [V(z_+, t) - V(z_-, t)] \hat{w} = 0. \quad (26)$$

While this does not yet look like Schrödinger's equation, this is a linear differential equation, and we can look at separation of variables to construct a general solution to (26). Suppose $\hat{w} = \sum \psi^+(z_+, t) \psi^-(z_-, t)$ where each of these terms satisfy the above, we have

$$\mp i\hbar \psi_t^\pm = -\frac{\hbar^2}{2M} \psi_{zz}^\pm + V(z_\pm, t) \psi^\pm. \quad (27)$$

Here z takes the respective roles of z_+ and z_- .

For real potential $V(x, t)$, $\psi^+(z_+, t) = \Psi^*(z_+, t)$ and $\psi^-(z_-, t) = \Psi(z_-, t)$, where $\Psi(x, t)$ is the original Schrödinger wavefunction (1).

If additionally $V(x, t)$ is time-independent, then $\psi^+ = e^{+i\lambda t} \psi_\lambda^*$ and $\psi^- = e^{-i\lambda t} \psi_\lambda$ where the ψ_λ are eigenvectors of the original Schrödinger's equation (1). These form a complete basis for the solution of PDE for $\hat{w}$:

$$\hat{w}(t, z_+, z_-) = \sum_{\lambda, \lambda'} C_{\lambda, \lambda'} e^{i(\lambda - \lambda')t} \psi_\lambda^*(z_+) \psi_{\lambda'}(z_-). \quad (28)$$

where the $C_{\lambda, \lambda'}$ are determined by the initial distribution w_0 . Assuming ψ_λ are normalized by $\int_{-\infty}^{+\infty} dz |\psi_\lambda(z)|^2 = 1$, we have:

$$C_{\lambda, \lambda'} = \int_{-\infty}^{+\infty} dz_+ \int_{-\infty}^{+\infty} dz_- \psi_\lambda(z_+) \psi_{\lambda'}^*(z_-) \hat{w}_0(z_+, z_-), \quad (29)$$

$$= \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} d\hat{p} \psi_\lambda(x + \frac{\hat{p}}{2}) \psi_{\lambda'}^*(x - \frac{\hat{p}}{2}) \hat{w}_0(x, \hat{p}). \quad (30)$$

Note C is Hermitian and w is normalized. This implies $C = \sum_k c_k a_k^\dagger a_k$ for real eigenvalues c_k , unit left eigenvectors a_k (so $a_k C = c_k a_k$), and $\sum_k c_k = 1$.

The quantity $w(x, p, t)$ is precisely the Wigner quasi-probability distribution [2], and $\hat{w}$ is its Fourier transform in p . In the quantum-to-classical direction, the exact evolution generated by $V(z_+, t) - V(z_-, t)$ and its expansion into the classical force term plus $O(\hbar^2)$ corrections are standard results [6, 5]. The looking-glass transform runs that arrow in reverse:

rather than starting from a wavefunction and constructing w , we started from the classical Liouville equation and found that a single centered difference converts its dynamics into Schrödinger dynamics. The mathematics of the correspondence is the classical Wigner–Moyal one; the contribution of this paper is the direction in which it is read, and what that reading buys.

Intuition. Quantum mechanics viewed through the lens of the Schrödinger equation carries essentially no classical intuition. Here is a partial dictionary between the two worlds. Any classical distribution evolves, to $O(\hbar^2)$, like a quantum-mechanical structure—though not every classical distribution is a quantum state; Section 8.1 returns to this. The reverse direction takes one additional mental leap: a uniform background probability. $w \rightarrow w + c$ for some constant c solves the local probability flux equation (19) just fine. This is the nod to quasi-probability. For many interesting Quantum configurations, the corresponding real quasi-probability,

$$w(x, p, t) = \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar} e^{+i\frac{\hat{p}p}{\hbar}} \Psi^*(x + \frac{\hat{p}}{2}, t) \Psi(x - \frac{\hat{p}}{2}, t) \quad (31)$$

is real-valued, integrates to 1, but not necessarily non-negative everywhere. Read it like a pressure perturbation about a uniform ambient level c : negative values are rarefactions, not missing probability, and classical intuition carries over (Section 8.4).

6 Error Estimate

The centered-difference approximation (14) is exact for linear and quadratic potentials. For other potentials, the leading error term is

$$\hat{p} \frac{\partial V}{\partial x} \approx V(x + \frac{\hat{p}}{2}) - V(x - \frac{\hat{p}}{2}) + \left[-\frac{\hat{p}^3}{24} \frac{\partial^3 V(x)}{\partial x^3} \right]. \quad (32)$$

For the approximation to be closer to exact, then the PDE (19) would have to be modified to

$$w_t + \frac{p}{M} w_x - \frac{\partial V}{\partial x} w_p = \left[-\frac{\hbar^2}{24} \frac{\partial^3 V}{\partial x^3} w_{ppp} \right]. \quad (33)$$

Read from the classical side, this is the correction that would make the centered-difference approximation exact; read from the quantum side, it is the leading Moyal correction [4, 15, 6] that separates Liouville from von Neumann dynamics. We note that the reading of this $\hbar^2$ term as the small parameter of a classical limit is not universally accepted: it has been argued that the correction series carries no explicit $\hbar$ and that the classical–quantum transition is

set by spatial scale rather than by $\hbar \rightarrow 0$ [16]. The structural statement below does not depend on which reading one adopts.

The error term is locally of order $\hbar^2$, but its effects are secular: in anharmonic potentials the classical and quantum evolutions diverge over long times—classically by filamentation of the phase-space distribution, quantum mechanically by interference, recurrences, and wave-packet revivals. Where the two dynamics part company, experiment follows the quantum side. The looking-glass transform therefore cannot support the claim that classical mechanics is fundamental and quantum mechanics merely its approximation. What it supports is the sharper structural statement that the *entire* difference between the dynamics of the two theories is the single term above.

7 The Born Projection

The standard Born rule states that at time t , the probability density of observing position x is $|\Psi(x, t)|^2$. We recover this as the instantaneous marginal of w over momentum, which we call the Born projection:

$$\text{born } w = \int_{-\infty}^{+\infty} dp w(x, p, t), \quad (34)$$

$$= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar} e^{i\frac{\hat{p}p}{\hbar}} \hat{w}, \quad (35)$$

$$= \int_{-\infty}^{+\infty} d\hat{p} \hat{w} \int_{-\infty}^{+\infty} \frac{dp}{2\pi\hbar} e^{i\frac{\hat{p}p}{\hbar}}, \quad (36)$$

$$= \int_{-\infty}^{+\infty} d\hat{p} \hat{w} \delta(\hat{p}), \quad (37)$$

$$= \hat{w}|_{\hat{p}=0}, \quad (38)$$

$$= \sum_{\lambda, \lambda'} C_{\lambda, \lambda'} e^{i(\lambda - \lambda')t} \psi_{\lambda}^*(x) \psi_{\lambda'}(x), \quad (39)$$

$$= \sum_k c_k \sum_{\lambda, \lambda'} (a_k)_{\lambda}^* (a_k)_{\lambda'} e^{i(\lambda - \lambda')t} \psi_{\lambda}^*(x) \psi_{\lambda'}(x), \quad (40)$$

$$= \sum_k c_k \left| \sum_{\lambda} (a_k)_{\lambda} e^{-i\lambda t} \psi_{\lambda}(x) \right|^2. \quad (41)$$

This is the standard Born position density, obtained here as the momentum marginal of the phase-space distribution. Marginal-based presentations of the Born densities are standard in the phase-space formulation [6, 17], and the claim that they are thereby *derived* appears in [7]. Within this representation the Born rule for position is not an additional

postulate: it is ordinary marginalization, and no collapse dynamics or decoherence mechanism [15] is invoked to obtain the statistics.

We are explicit about the limits of this deflation. It reproduces the statistics of position (and, by the symmetric argument, momentum) measurements. It does not by itself explain definite individual outcomes or the statistics of observables that are not phase-space marginals—those require overlaps $\int w W_A$ with Wigner symbols W_A that themselves take negative values. The update of w conditioned on an observed result, for measurements that are marginals, is treated in Section 8.3. The remaining gaps are collected in Section 9.

8 Interpretive Remarks

8.1 Heisenberg’s Uncertainty Principle

Within the looking-glass dynamics there is nothing inconsistent about a distribution that specifies the precise initial position and momentum of the particle; the equations evolve it without complaint. However, we can only observe the marginal over momentum, which makes the view fuzzy.

Substitution of $w_0(x, p) = \delta(x - x_0)\delta(p - p_0)$ into the formula for $C_{\lambda, \lambda'}$ gives the cross-Wigner function as the coefficients of the expansion of $\hat{w}$ in the eigenbasis of the Schrödinger equation:

$$C_{\lambda, \lambda'}[\delta(x - x_0)\delta(p - p_0)] = \int_{-\infty}^{+\infty} d\hat{p} \psi_{\lambda}(x_0 + \frac{\hat{p}}{2}) \psi_{\lambda'}^*(x_0 - \frac{\hat{p}}{2}) e^{-i\frac{\hat{p}p_0}{\hbar}}. \quad (42)$$

For a symmetric potential V , the eigenfunctions ψ_{λ} are either symmetric or antisymmetric, $\psi_{\lambda}(-x) = \sigma_{\lambda}\psi_{\lambda}(x)$, where $\sigma_{\lambda} = \pm 1$. We can use this to compute $C_{\lambda, \lambda'}$ for the $x_0 = 0$, $p_0 = 0$ case:

$$C_{\lambda, \lambda'}[\delta(x)\delta(p)] = \int_{-\infty}^{+\infty} d\hat{p} \psi_{\lambda}(\frac{\hat{p}}{2}) \psi_{\lambda'}^*(-\frac{\hat{p}}{2}), \quad (43)$$

$$= \sigma_{\lambda'} \int_{-\infty}^{+\infty} d\hat{p} \psi_{\lambda}(\frac{\hat{p}}{2}) \psi_{\lambda'}^*(\frac{\hat{p}}{2}), \quad (44)$$

$$= 2\sigma_{\lambda'} \int_{-\infty}^{+\infty} du \psi_{\lambda}(u) \psi_{\lambda'}^*(u), \quad (45)$$

$$= 2\sigma_{\lambda} \delta_{\lambda, \lambda'}. \quad (46)$$

The Born projection yields

$$\text{born } \delta(x)\delta(p) = 2 \sum_{\lambda} \sigma_{\lambda} |\psi_{\lambda}(x)|^2. \quad (47)$$

This is the expansion of $\delta(x)$ as a distribution—what we would expect for a particle at the bottom of the well with no momentum.

Two caveats sharpen this picture. First, the coefficient matrix C of the point preparation is Hermitian with unit trace but *not* positive semidefinite: the eigenvalues $c_k = 2\sigma_{\lambda}$ alternate in sign. A point in phase space is therefore not a quantum state—necessary and sufficient conditions for a phase-space function to be a Wigner distribution are known [12]—and the uncertainty principle reappears in this representation as the positivity condition on C , not merely as fuzziness of the marginals. Second, the looking-glass dynamics is nevertheless well posed on the larger set of Hermitian unit-trace C , and positivity is preserved by the evolution, so the quantum states form an invariant subset. Classical point preparations live in the complement: they evolve consistently and project to sensible signed densities, as above, but no quantum preparation reaches them. What principle selects the positive subset is an open question we return to in Section 9.

8.2 Quantum Entanglement

The above derivation is for a single particle in a single space dimension, but it is trivial to extend to multiple particles in three spatial dimensions. Entanglement amounts to a dependent choice of initial distribution, so $\hat{w}$ cannot be factored into independent components for each particle. The underlying mechanism evolves independently; it is only the initial distribution and the momentum marginal that make it look like there is something mysterious. It is a consequence of dependent distributions and the projection.

For two particles with single-particle eigenstates $\hat{\alpha} = \psi_{\lambda}$ and $\hat{\beta} = \psi_{\mu}$, the coefficient matrix C extends to the two-particle eigenbasis $\{|\alpha\alpha\rangle, |\alpha\beta\rangle, |\beta\alpha\rangle, |\beta\beta\rangle\}$. The four maximally entangled Bell states are pure states $C = a a^{\dagger}$ with

$$a_{\Phi\pm} = \frac{1}{\sqrt{2}}(|\alpha\alpha\rangle \pm |\beta\beta\rangle), \quad (48)$$

$$a_{\Psi\pm} = \frac{1}{\sqrt{2}}(|\alpha\beta\rangle \pm |\beta\alpha\rangle). \quad (49)$$

None of these factor as $|s_1\rangle \otimes |s_2\rangle$; entanglement is precisely this non-factorizability of a . The outer product $a a^{\dagger}$ generates off-diagonal coherence terms in $\hat{w}$ that distinguish entangled states from classical mixtures, which would have C diagonal.

This demystification of entanglement—as non-factorizable statistical dependence in the

joint initial data—is not new in spirit: related readings appear in Wetterich’s derivation of entanglement from classical statistics [18] and in Ozorio de Almeida’s representation of an entangled state as a non-factorizable Lagrangian surface in phase space [19]. It also has a sharp limit that must be stated. A Bell-type entangled state admits *no* non-negative joint phase-space distribution reproducing all of its measurement statistics; this is a finite, explicit no-go [20, 21]. The dependent object here is therefore the signed w of Section 8.4, not an ordinary non-negative probability: entanglement is statistical dependence of initial data in that signed field. The obstruction is moreover observable-dependent—a positive-Wigner Gaussian state still violates a Bell inequality when displaced-parity observables are measured [22], precisely because those observables are not phase-space marginals. The remark above thus demystifies the *kinematics* of entanglement (non-factorizable C); measurement and conditioning are taken up next.

8.3 Collapse of a Separated Entangled Pair

The kinematic reading above leaves the sharpest question open: what happens to A when a distant B is measured? We model measurement in the von Neumann manner [23], as interaction with an apparatus, and take the interaction to involve B alone. For this question no relativity is needed, and the particles may be free.

Every step is exact. The error term of Section 6 involves $\partial^3 V / \partial x^3$, and more generally the Moyal correction vanishes for any Hamiltonian at most quadratic in the phase-space variables. For two free particles the looking-glass dynamics is therefore exactly the classical free streaming of the joint distribution,

$$w(x_A, p_A, x_B, p_B, t) = w_0\left(x_A - \frac{p_A t}{M}, p_A, x_B - \frac{p_B t}{M}, p_B\right), \quad (50)$$

and nothing quantum happens dynamically: everything quantum about the pair resides in w_0 , in its correlations and its sign. Couple B to a pointer with position X and momentum P by an impulsive interaction $H_{\text{int}} = g(t) x_B P$ of integrated strength g . It is bilinear, so the centered difference remains exact, and Hamilton’s equations give

$$X \rightarrow X + g x_B, \quad p_B \rightarrow p_B - g P : \quad (51)$$

the pointer records B 's position, and B receives a momentum kick. Liouville flow carries the joint distribution to

$$w_{\text{after}} = w_{AB}(x_A, p_A, x_B, p_B + gP) w_{\text{ptr}}(X - gx_B, P). \quad (52)$$

The interaction involves only B and its apparatus; integrating out B and the pointer leaves A 's marginal unchanged. The flow is local.

Collapse is conditioning. Reading the pointer is taking its X -marginal, the same operation as the Born projection. Conditioning on the reading $X = X_0$ and integrating out P and p_B —which removes the kick—gives

$$w_A(x_A, p_A | X_0) \propto \int dx_B \rho_{\text{ptr}}(X_0 - gx_B) \int dp_B w_{AB}(x_A, p_A, x_B, p_B), \quad (53)$$

where ρ_{ptr} is the pointer's initial position density. This is exactly the Wigner function of A 's quantum state after an unsharp position measurement on B ; in the sharp-pointer limit it reduces to the standard identity that the Wigner function of $\langle x_B | \hat{\rho} | x_B \rangle$ is the joint w marginalized over p_B and evaluated at x_B . Averaging (53) over readings returns A 's original marginal: nothing done at B signals to A . The instantaneous change in A 's description is a change of conditioning, not a change of dynamics. Within this representation, the “collapse” of a separated entangled pair is conditioning of a joint distribution on the marginal of a pointer that interacted only with B .

Where the quantum ingredient lives. If $w_{AB} \geq 0$, (53) is ordinary classical conditioning; Bell made exactly this point for the original EPR state, whose Wigner function is non-negative and is itself a local hidden-variable model [13]. If w_{AB} is signed, (53) is formally identical but is Bayes' rule with signed weights. Different measurement choices—measuring x_B at time t is measuring the quadrature $x_B - p_B t/M$ at time 0—are different marginals of the same signed w , and no single non-negative distribution reproduces them all; non-Gaussian states violate Bell inequalities with such quadrature measurements alone [24, 25]. The conditional is a signed field divided by a positive marginal probability, so it is an object of the same kind as w itself, and its readings are again non-negative marginals.

The size of the apparatus. An apparatus of N pointer modes, each coupled as $g x_B P_k$ with initial width σ , reduces exactly to one collective pointer: $\bar{X} = N^{-1} \sum_k X_k$ and $P_{\text{tot}} = \sum_k P_k$ are conjugate, the coupling involves only P_{tot} , and the $N - 1$ relative modes decouple. The collective pointer is a minimum-uncertainty Gaussian of width $\sigma/\sqrt{N}$, so a large

apparatus is one whose collective reading is sharp even when no single mode can resolve B . The price is the uncertainty principle as back-action: the kick gP_{tot} has variance $Ng^2\hbar^2/4\sigma^2$. In the momentum distribution this kick is a convolution, and an A – B interference term oscillating in p_B at the conjugate separation Δx_B is suppressed by

$$\exp\left(-\frac{Ng^2\Delta x_B^2}{8\sigma^2}\right). \quad (54)$$

In the larger state space of Section 8.1, a point pointer $\delta(X)\delta(P)$ would measure B perfectly without disturbing it; the positivity condition on the pointer is what forces the back-action.

Numerical illustration. We take $\hbar = M = 1$, Gaussian packets $\phi_c(x)$ of position width 0.5 centered at c , and the entangled state

$$\Psi_{AB} = \alpha |\text{cat}_+\rangle_A |L\rangle_B + \beta |\text{cat}_-\rangle_A |R\rangle_B, \quad |\alpha|^2 = 0.7, \quad (55)$$

with $\text{cat}_\pm \propto \phi_{-2} \pm \phi_{+2}$ and $L, R = \phi_{\mp 2}$. The apparatus couples only to B with $g = 1$; a single pointer of width $\sigma = 6$ cannot resolve B 's branches, which are $\Delta x_B = 4$ apart. The classical calculation never uses a wavefunction after constructing w_0 : it pushes signed phase-space weights through the Hamiltonian flow above, reads the collective pointer, and conditions on the sign of the reading r . An independent quantum calculation evolves $\Psi(x_A, x_B, X)$ on a grid with $\exp(-igx_B\hat{P}/\hbar)$ and projects. The code and the environment that runs it accompany the paper.

N	$P(r > 0)$		A – B coherence		$\min w_A(\cdot r > 0)$	
	classical	quantum	classical	quantum	classical	quantum
1	0.448	0.446	0.96	0.95	−0.040	−0.036
4	0.403	0.400	0.81	0.80	−0.035	−0.036
16	0.342	0.340	0.41	0.41	−0.185	−0.185
64	0.306	0.305	0.03	0.03	−0.300	−0.299
256	0.300	0.300	0.02	0.00	−0.318	−0.317

Table 1: Classical (signed Liouville flow plus conditioning) against quantum results. The coherence column for (54) reads 0.95, 0.80, 0.41, 0.03, 0.00; the classical floor near 0.02 is Monte Carlo noise from the signed weights.

Figures 1 and 2 and Table 1 show five features. The probability of each reading approaches the Born weight as the apparatus grows. A 's unconditional distribution is unchanged by the measurement, and the probability-weighted average of the two conditionals reproduces it to machine precision. Conditioning selects A 's branch: given $r < 0$, A is cat_+ , and given $r > 0$, it is cat_- . The apparatus destroys the A – B coherence at the rate (54), after

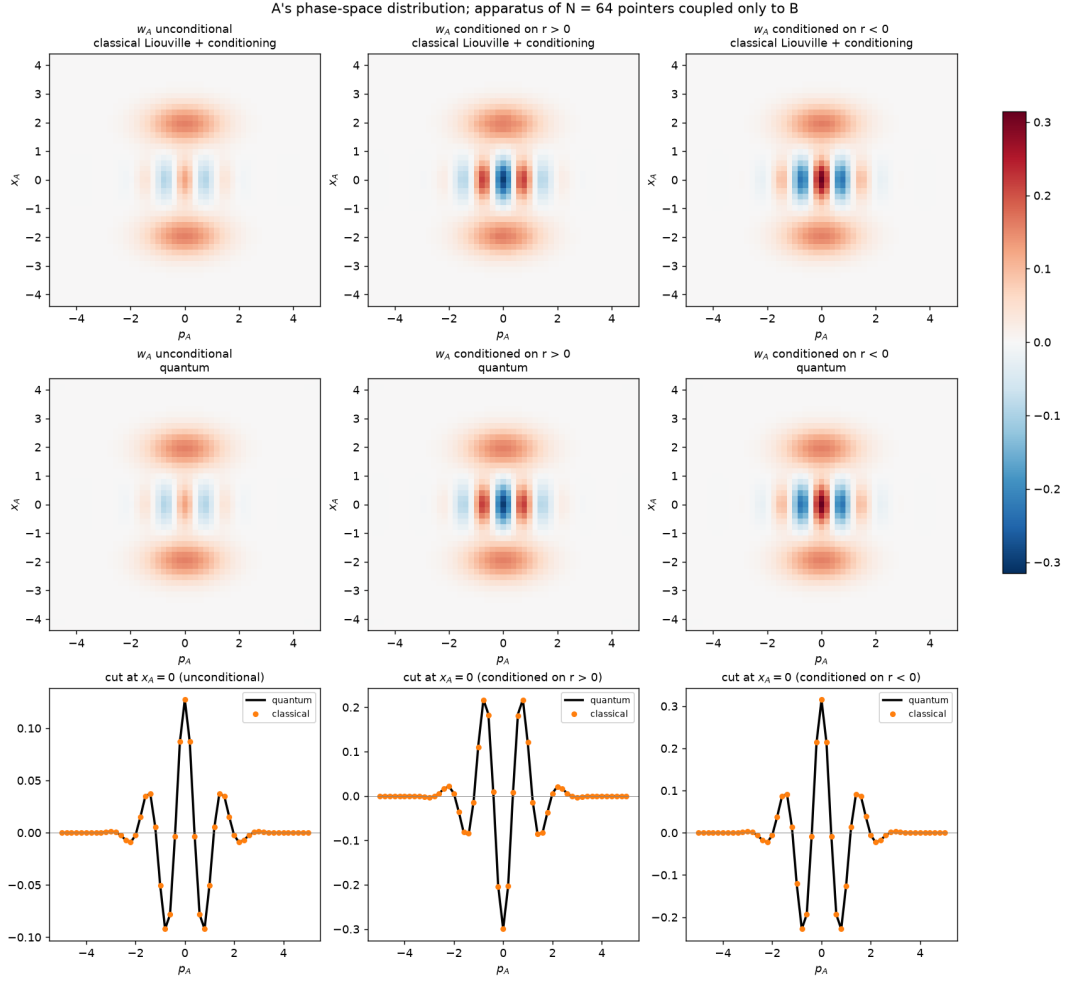


Figure 1: A 's phase-space distribution for $N = 64$ pointers coupled only to B . Top row: classical Liouville flow of the signed w_{AB} followed by conditioning on the pointer reading. Middle row: independent quantum calculation. Bottom row: cuts at $x_A = 0$ (line: quantum; dots: classical). Conditioning on $r < 0$ or $r > 0$ selects cat_+ or cat_- , with $w_A(0, 0) = \pm 1/\pi\hbar$, the bound of Section 8.4.

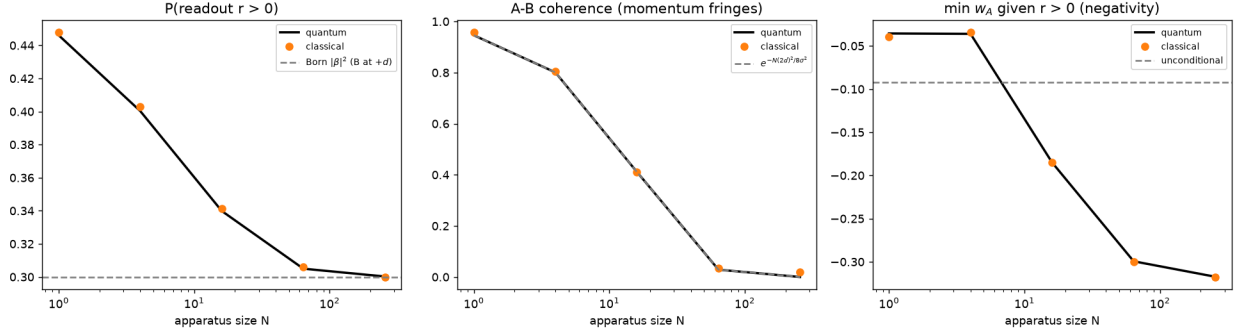


Figure 2: Dependence on apparatus size N . Left: probability of the reading $r > 0$, approaching the Born weight $|\beta|^2 = 0.3$. Center: A – B coherence, measured by the interference fringe of the joint momentum distribution, against (54). Right: minimum of A ’s conditional distribution. Line: quantum; dots: classical.

which the joint momentum distribution is indistinguishable from a classical mixture of the branches. And conditioning *creates* negativity: A ’s minimum falls from -0.09 unconditionally to $-1/\pi\hbar \approx -0.318$ given the reading, which conditioning with non-negative weights could never do.

Three limits should be stated. First, because every Hamiltonian here is quadratic, agreement between the classical and quantum calculations is guaranteed; it verifies the construction and the code, not a prediction that could have failed. The content is the reading—local flow, collapse as conditioning, Born weights from a pointer marginal—not the agreement. Second, “large” enters only through the sharpness of the collective reading; the record here is reversible in principle, and amplification into an irreversible record is not modeled. Third, the argument covers measurements that are phase-space marginals; observables that are not (Section 9) are outside it.

8.4 Negative Values of w

Every reader meeting a phase-space distribution that dips below zero asks the same question, so we answer it directly: the negative values do not matter, for two independent reasons.

w is a field, like a pressure wave. The Liouville equation (19) is a statement about *flux*, not about absolute values. If w satisfies it, so does $w + c$ for any constant c : a uniform background has zero gradient and carries no flux. An acoustic pressure wave is the familiar example. The pressure perturbation is negative in every rarefaction, and nobody objects, because what is physical is the total pressure, ambient level plus perturbation, and only gradients drive the flow. The same holds here. For every pure state the Wigner function

obeys

$$-\frac{1}{\pi\hbar} \leq w(x, p, t) \leq \frac{1}{\pi\hbar}, \quad (56)$$

so $w = \rho - c$ with $\rho \geq 0$ and the universal ambient level $c = 1/(\pi\hbar)$, one count per quantum cell. The negative regions of w are rarefactions below that level, not places where probability is missing. Like the ambient pressure of an unbounded medium, the background has no finite total, and it needs none: it carries no dynamics and enters no prediction. We have not found this reading in the negativity literature, where negativity is usually treated as the irreducibly quantum feature [11].

A correct model owes us nothing more. A model is judged by its predictions, and every prediction made here is non-negative. The Born projection born $w = \int dp w = |\Psi(x, t)|^2$ and the momentum marginal $\int dx w = |\tilde{\Psi}(p, t)|^2$ are genuine probability densities whatever the sign of w , as are the pointer readings of Section 8.3. Intermediate quantities in a calculation need not be probabilities; Feynman made this argument for negative probabilities in general [26]. Where the error term of Section 6 is negligible, the model is correct, and that is the only test that applies to it. Where it is not, experiment follows the quantum dynamics; that is the real limit of the framing, and it has nothing to do with sign.

What the negative regions mark. Negative regions have been measured directly by quantum state tomography [27]. By Hudson’s theorem [28], $w \geq 0$ everywhere exactly for Gaussian states, so the rarefactions mark the states with no Gaussian description, such as Fock states and superpositions. They are also where tunneling and interference show up in phase space; those effects are carried by the retained Schrödinger terms, not by the Moyal correction that is dropped for anharmonic potentials.

8.5 Many Worlds

The many-worlds interpretation [1] posits branching universes to resolve the measurement problem: if the wavefunction never collapses, what selects the observed outcome? The present framework removes one motivation for branching: the Born statistics arise by marginalization, with no collapse event anywhere in the dynamics. The apparent indeterminacy of quantum mechanics enters through the Born projection—the momentum marginal—which discards momentum information that is inaccessible to a position measurement, identical in form to how a classical observer recovers a position probability from an (x, p) distribution. This does not by itself resolve the measurement problem—Section 7 lists what marginalization leaves unexplained—but it relocates the apparent indeterminacy from the dynamics to

the projection, where classical intuition can engage it.

9 Further Directions

The framing leaves honest gaps, each a direction for further work.

9.1 Spin and statistics

The construction covers spinless particles. Antisymmetric a vanishes for coincident single-particle states, so Pauli exclusion is representable, but nothing in the transform explains why identical particles should carry symmetric or antisymmetric C ; spin and the spin-statistics connection are absent.

9.2 General observables

The Born densities for position and momentum are marginals of w . A general observable A requires the overlap $\int dx dp w W_A$, where the Wigner symbol W_A itself takes negative values, and the clean reading of measurement as marginalization is then lost. Whether the looking-glass framing offers any purchase on general measurements is open.

9.3 The larger state space

The dynamics is well posed for every Hermitian, unit-trace C ; positivity is preserved by the evolution; and the quantum states form an invariant subset whose complement contains all classical point preparations. Three questions follow. What physical principle selects the positive subset? Do non-positive C —classically preparable in this representation—have any operational meaning, or are they strictly bookkeeping? And does the boundary of the quantum subset, where some $c_k = 0$, play a distinguished dynamical role? The conditions of [12] give the technical footing for these questions.

9.4 The uniform background

The ambient level $c = 1/(\pi\hbar)$ makes $\rho = w + c \geq 0$ for every pure state. Like the ambient pressure of an unbounded medium it carries no dynamics; whether it acquires a physical role in a finite box, or in field theory as vacuum fluctuation, is unexplored.

9.5 Pedagogy

The route from Liouville to Schrödinger requires only Fourier transforms and centered differences—tools familiar to any student of numerical methods. A classroom treatment with numerical experiments (exactness for the harmonic oscillator, visible error growth for quartic potentials, revivals versus filamentation) would test whether the looking-glass framing actually transfers classical intuition to the quantum domain.

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